

TEE-RPCH Experimental Preview on MNT224

Anonymous
Anonymous Institution

ABSTRACT

This preview document isolates the MNT224 experimental results generated from the completed real-SGX measurements. The preview uses the full data_mnt224 set for baseline, hash-check, and concurrency plots, and the refreshed targeted_tables/raw/mnt224 revocation results for the revocation section.

1 EXPERIMENTAL SCOPE

The preview focuses on the corrected five-experiment structure: a theory-level comparison, a secure baseline comparison, the user-side hash-check benefit, a multi-thread experiment, and a revocation comparison. All TEE-RPCH results in this preview come from the completed MNT224 hardware-SGX run.

2 EXPERIMENT 1: THEORY COMPARISON

3 EXPERIMENT 2: BASELINE COMPARISON

4 EXPERIMENT 3: HASH-CHECK BENEFIT

5 EXPERIMENT 4: MULTI-THREAD EVALUATION

6 EXPERIMENT 5: REVOCATION

Table 1: Theory comparison.

scheme	keygen	hash	adapt
FB-PCH	$O(S)$	$O(policy)$	$O(policy)$
EHR-RPCH	$O(S + \log N)$	$O(policy)$	$O(policy + \text{outsourced revoke})$
PNITCH	$O(S)$	$O(1)$	$O(t)$ partial adapt + combine
RPCH-XNM'21	$O(S + \log N)$	$O(policy)$	$O(policy + \log N)$
TAR-PCH	$O(S + \log N)$	$O(policy)$	$O(policy + \text{cloud update})$
TPCH	$O(S)$	$O(policy)$	$O(policy + \text{update rounds})$
TEE-RPCH (ours)	$O(S)$	$O(policy)$	Server $O(policy)$, TEE $O(\log N)$, User $O(1)$

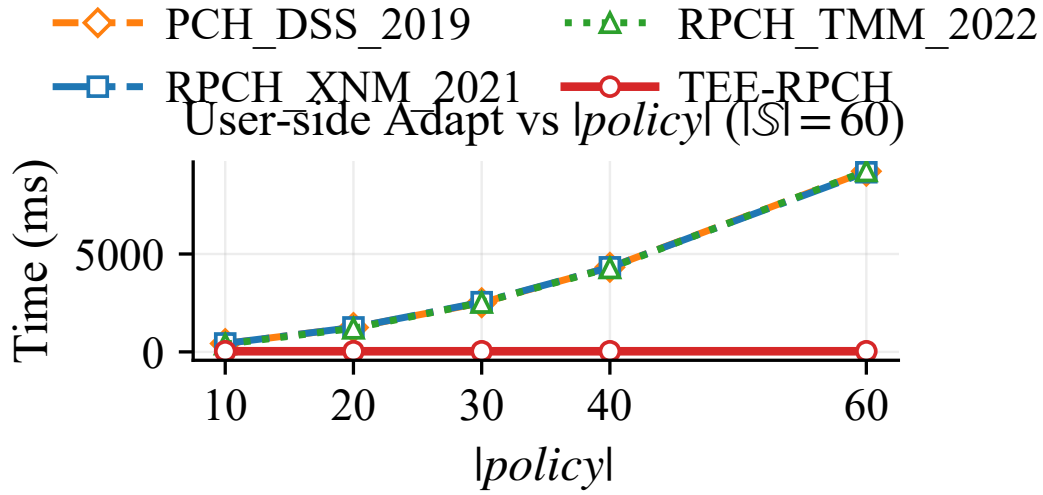


Figure 1: Online adaptation cost versus policy size.

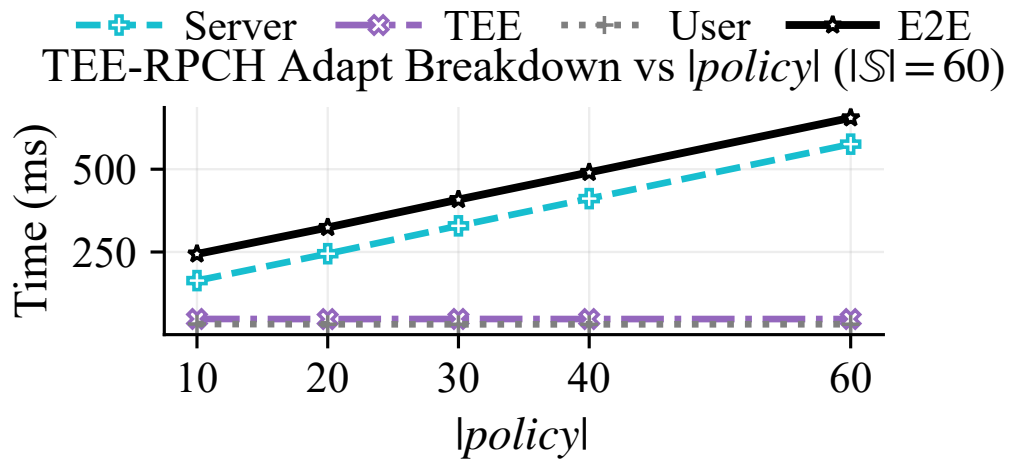


Figure 2: Server, enclave, and user breakdown.

Table 2: MNT224 baseline comparison (ms).

policy	single_outsource_ms	tee_rpch_e2e_ms	server_ms	tee_ms	user_ms
10	98.977	243.907	163.597	47.774	32.536
20	139.716	324.002	244.844	47.390	31.768
30	180.884	408.361	329.024	47.719	31.618
40	225.630	490.304	411.217	47.496	31.591
60	305.502	655.117	575.734	47.510	31.873

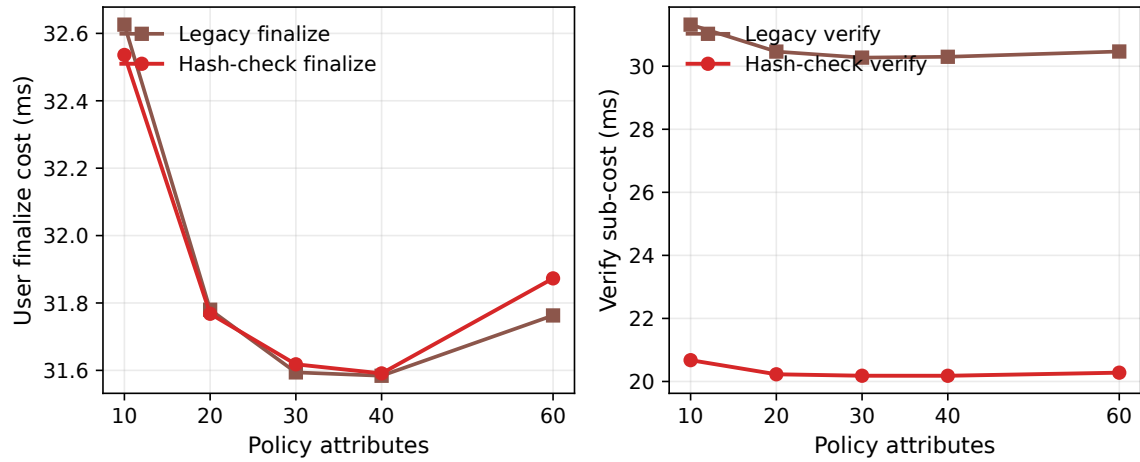


Figure 3: Legacy finalize path versus hash-check path on MNT224.

Table 3: MNT224 user-side finalize comparison (ms).

policy	legacy_user_ms	hashcheck_user_ms	legacy_verify_ms	hashcheck_verify_ms
10	32.626	32.536	31.319	20.676
20	31.780	31.768	30.464	20.228
30	31.594	31.618	30.273	20.182
40	31.584	31.591	30.299	20.181
60	31.763	31.873	30.467	20.279

Table 4: MNT224 revocation comparison (ms).

users	ehr_kgc_ms	ehr_csp_ms	ehr_total_ms	tar_aa_ms	tar_csp_ms	tar_total_ms	tee_kgc_ms	tee_ms	tee_total_ms
2 ¹ 0	12.047	2192.210	2204.260	1697.620	2321.130	4018.750	0.042	0.158	0.200
2 ¹ 2	12.406	2173.490	2185.900	3720.340	2324.670	6045.020	0.043	0.158	0.201
2 ¹ 4	13.201	2203.820	2217.020	11785.500	2342.120	14127.600	0.049	0.174	0.223
2 ¹ 6	16.973	2202.070	2219.040	43885.700	2330.190	46215.900	0.059	0.206	0.265
2 ¹ 8	29.015	2193.030	2222.040	172162.000	2338.860	174501.000	0.070	0.233	0.303

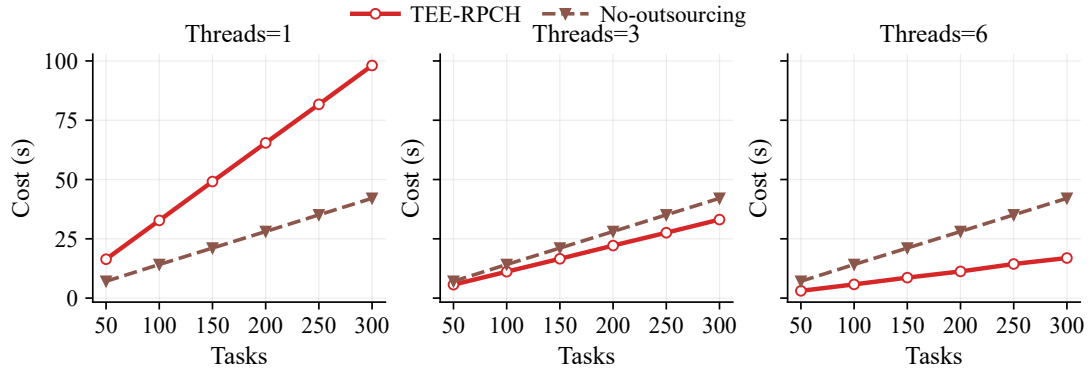


Figure 4: Wall-clock cost versus task count.

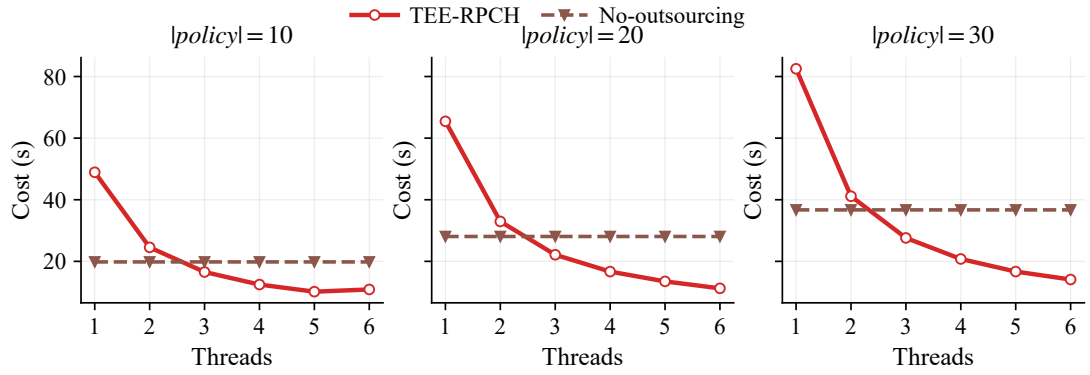


Figure 5: Wall-clock cost versus server thread count.

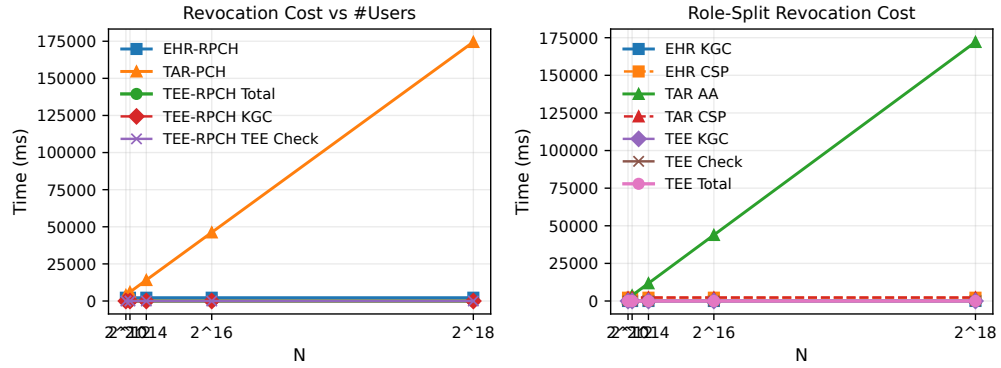


Figure 6: Revocation cost versus number of users.